Supplementary Methods and Source-Integrity Audit

Resolution Visibility Is Not Recovery (Public-reproducibility update)

Author review copy; authorship pending confirmation

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Status and scope. This technical supplement documents deterministic reanalysis of a public annotation file and post-hoc descriptive stability checks. It is *not* independent label validation, a human-coder study, a systematic literature sample, or an estimate of causal severity effects. The original author-annotated rows are not redistributed.

S1. Frozen data, provenance, and admission gates

The unique source is the public annotation object supporting Tang et al., *How Coding Agents Fail Their Users: A Large-Scale Analysis of Developer–Agent Misalignment in 20,574 Real-World Sessions* (EMNLP 2026; DOI 10.18653/2026.emnlp-main.1011). Our calculation uses the fixed code-release commit and path: **Pinned GitHub source file**.
Commit: `c887cc920129b2db0bdf218c8550e98f74f42f7c`.
Path: `data/core/annotated_misalignments.json`.
Source SHA-256:
`7e0693675242a625bdb586bcd8cef23f5da76c65cf840108c3c24b6b479b9f57`.

The source has 16,118 episode objects. Each composite episode key (source, repository, session, misalignment) is distinct. The observed resolver labels agree structurally with resolution status: the 1,504 records marked RS1 have one RV1/RV2/RV3 resolver; the 14,614 records marked RS2 have null resolver. The source creates its own severity and resolution labels; no reannotation is performed here. Its observational selection rule requires a mismatch made visible by developer correction or pushback, rather than sampling all failures. The source's arXiv v1 (28 May 2026), Appendix A.2, Table 5, reports on 100 adjudicated records: damage-severity agreement between two source annotators was 0.71, and model-to-gold accuracy was 0.64; the respective resolution-status values were 0.92 and 0.96. These are source-reported per-axis exact-match diagnostics, not a known DS1/DS2 confusion matrix, a full-corpus correction rate, or our own validation. The accepted EMNLP version and released data are distinguished from this archived annotation-quality edition.

Table 1: Preserved source partitions. RS2 is unknown, not failure.

Subset	Episodes	Visible RS1	Visible RV2
DS1 (effort/trust only)	14,587	1,174	1,080
DS2 (easily reversible damage)	1,361	330	296
Other severity classes (DS0/DS3/DS4)	170	0	0
Total	16,118	1,504	1,376

Observation–classification coupling. The original codebook requires evidence of an applied, reversible code/state change for DS2, while RS1 requires visible confirmation of correction. Both labels are inferred from the same textual record. This leaves open common-visibility and classification mechanisms even after repository/session matching; the source provides no linked dual-annotator DS1/DS2 confusion matrix to separate them.

S2. Units, weights and dependence

Our primary descriptive endpoint is $Y_i = \mathbf{1}\{\text{resolution_status}_i = \text{RS1}\}$, where $Y = 0$ means no recorded visible resolution rather than known failure. In a session s containing both classes, define $D_s = \bar{Y}_{\text{DS2},s} -$

$\bar{Y}_{\mathrm{DS1},s}$. The equal-session estimator is $\bar{D}_S = |S|^{-1} \sum_{s \in S} D_s$. There are 853 paired sessions with 2,775 eligible episodes, represented by 352 distinct source–repository keys. The equal-session estimate is 0.1396122927.

To preserve clustering beyond the session level, the MR22 descriptive bootstrap selects 352 source–repository clusters with replacement, keeping *all* matched sessions within each draw. The equal-session mean is recomputed over the resampled sessions (3,000 draws; fixed seed 993720); the 2.5th and 97.5th nearest-order percentiles are $[0.1071711681, 0.1729976896]$. The older whole-session resampling interval, $[0.1090339418, 0.1702394909]$ (seed 993719), is shown only for comparison because sessions are nested in repositories. The alternative equal-repository estimator gives each of the 352 repository means of session differences equal weight: 0.1405407792, with repository-resampling percentiles $[0.1017798217, 0.1809355907]$.

Table 2: MR22 robustness endpoints (all values in percentage points). Each interval is a descriptive resampling band for this observed nonrandom file.

Restriction/weighting	Sessions	Repos	Mean	Repo-level 2.5–97.5 band
All paired sessions, equal session	853	352	+13.96	+10.72 to +17.30
All paired sessions, equal repository	853	352	+14.05	+10.18 to +18.09
Extraction confidence = high	801	339	+14.30	+10.87 to +17.66
Exactly one episode of each class	307	171	+16.94	+11.73 to +22.36

Removing each repository in turn from the 853-session calculation yields an equal-session mean in $[0.1355586404, 0.1435967878]$. In the matched sessions, the DS2 rate exceeds the DS1 rate in 201 sessions, is lower in 99, and ties in 553. The original `confidence` attribute measures support for episode extraction, not certainty of damage-severity coding. Subset comparisons were chosen *after* viewing available fields, are exploratory, and may select a nonrepresentative fraction of the file. No adjustment for an exploratory search over many candidates is claimed; no inferential p-values are reported.

S3. Logical bounds and assumption frontier

Let r_g be the visible RV2 share among all episodes in severity group g, and u_g the fraction with unknown final status. Define $q_g \in [0, 1]$ as a *hypothetical* eventual RV2 fraction among currently unknown cases. Then

$$\theta_g = r_g + u_g q_g, \qquad \Delta = \theta_{\mathrm{DS2}} - \theta_{\mathrm{DS1}} = \left(\frac{296 + 1031 q_2}{1361}\right) - \left(\frac{1080 + 13413 q_1}{14587}\right).$$

Without assumptions, $\Delta \in [-0.7760687641, 0.9009798414]$; the endpoints are logically sharp completions of currently unknown cases, not probabilistic uncertainty limits. The exact zero-difference frontier has the form

$$q_1 = \frac{296/1361 - 1080/14587}{13413/14587} + \frac{1031/1361}{13413/14587} q_2 \approx 0.1560 + 0.8238 q_2.$$

Table 3: Illustrative *assumption* thresholds, not observed recovery estimates.

q_2: unknown DS2 RV2 share	q_1 threshold for a tie
0%	15.60%
25%	36.20%
50%	56.79%
75%	77.39%
100%	97.98%

The special restriction $q_1 = q_2 = q$ gives $q = 0.8856$ as a sign crossing. Without a defensible restriction on q_1, q_2, the sign of eventual recovery is not identified. The plotted line in the main manuscript visualizes this algebra; it is not a fitted empirical curve.

S4. Worst-case severity-reclassification certificate

This check is separate from the missing-eventual-outcome model in Section S3. We hold the observed binary RS1/RS2 outcomes fixed and allow the source severity label to move between DS1 and DS2 (no DS0/DS3/DS4 changes). The observed 2-by-2 table is DS1 (RS1=1,174; RS2=13,413) and DS2 (RS1=330;

RS2=1,031). To lower the observed DS2-minus-DS1 difference, it can be beneficial to move an RS1 from DS2 to DS1 (count x) or an RS2 from DS1 to DS2 (count y). Moves in the reverse directions only increase the positive contrast until crossing. Under the allowed changes, the modified comparison is

$$\frac{330 - x}{1361 - x + y} \;\le\; \frac{1174 + x}{14587 + x - y}, \quad 0 \le x \le 330, \quad 0 \le y \le 13413.$$

We enumerate integer $k = x + y = 0, 1, \ldots$ and all feasible pairs $0 \le x \le k$ using exact cross-multiplication, stopping at the first non-positive difference. The first witness occurs at $k = 223$, $(x, y) = (223, 0)$; all $k < 223$ fail the inequality. The witness produces 107/1,138 (9.4025%) versus 1,397/14,810 (9.4328%), respectively, and alters 223/15,948 (1.3983%) original two-class labels. Restricting the same exercise to each acquisition source separately yields the following exact minima:

Stratum	DS1 visible/N	DS2 visible/N	Min. flips	Fraction of eligible labels
Pooled	1,174/14,587	330/1,361	223	1.3983%
SpecStory	833/10,861	255/1,107	170	1.4205%
SWE	341/3,726	75/254	55	1.3819%

The source validation sample does not tell us whether even one such targeted flip is actually warranted. Consequently these minima are *worst-case algebraic perturbation budgets, not estimated annotation error rates, not adjusted results, and not confidence limits.* The observed endpoint depends on classification validity; the unknown eventual recovery endpoint remains separately unidentified even if every severity label were flawless.

S5. Reproduction and claim gates

Run the scripts with Python 3 (standard library only), after obtaining `annotated_misalignments.json` from the pinned author source. Use the exact source file above; the script refuses any other SHA-256. In the Overleaf source ZIP, execute:

```
python reproduce_visibility_robustness.py \
  --annotations /path/to/annotated_misalignments.json \
  --output MR21_check.json
python reproduce_nested_robustness.py \
  --annotations /path/to/annotated_misalignments.json \
  --output MR22_check.json
```

The JSON output is aggregate-only and includes a deterministic integer-search certificate for minimum severity relabeling budgets. The script enumerates all outcome-concordant DS1/DS2 switches, verifies each candidate by exact integer cross-products, and keeps RS1/RS2 fixed. The complete source records are never copied into the public manuscript or supplementary ZIP. The post-hoc checks are explicitly marked exploratory. Assertions enforce the frozen input, source partitions, 414 mixed-label repositories, 853 mixed-label sessions, 352 repositories supporting those sessions, 801 high-confidence mixed-label sessions, and 307 one-to-one mixed-label sessions.

Interpretive holds. Neither a repository bootstrap nor matching across recorded sessions can validate the authors' original DS1/DS2 classification, determine RS2 terminal outcomes, or support causal intervention estimates. The original source's 91.49% post-pushback resolution proportion is valid *only conditional on the visibly resolved 1,504 episodes.* Source-reported validation on the severity axis is materially weaker than on the resolution-status axis in the explicitly identified May 2026 arXiv v1, and no new human reannotation is claimed in this work. Publisher template updates, authorship, CRediT, funding, conflict declaration, exact AI-use disclosure and author-approved persistent public deposit remain separate administrative checks.